MALL CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING
Project: Mall Customer Segmentation
Application Framework: Streamlit
Algorithm: K-Means Clustering

1. Introduction

Customer segmentation is a machine learning application used to group customers according to similarities in their characteristics and behaviour. Businesses can use these groups to better understand their customers and support marketing and customer engagement decisions.

This project implements a Mall Customer Segmentation system using the K-Means clustering algorithm. The model groups customers based on annual income and spending score. A Streamlit application provides an interactive interface through which users can enter customer information and receive a predicted cluster and its associated segment name.

2. Project Objectives

The main objectives of the project are:

- To explore and prepare customer data for clustering.
- To apply the K-Means clustering algorithm.
- To group customers according to annual income and spending score.
- To save the trained machine learning pipeline.
- To develop an interactive application using Streamlit.
- To test the application with multiple customer inputs.

3. Dataset and Features

The model uses two numerical input features.

Annual Income (k$): Represents annual income in thousands of dollars. For example, a value of 50 represents an annual income of $50,000.

Spending Score (1–100): Represents a customer's spending score on a scale from 1 to 100.

These features are used to determine the similarity between customers during clustering.

4. Methodology

The project follows a machine learning workflow consisting of data understanding, data preparation, model development, evaluation, model saving, and application deployment.

4.1 Data Preparation

The dataset is explored and prepared for clustering. The selected features are annual income and spending score. Data preparation should ensure that these features are in the correct numerical format and suitable for the selected algorithm.

4.2 Model Development

K-Means clustering is used to group customers into five clusters. The pipeline observed during application testing contains a StandardScaler followed by a K-Means model with five clusters, ten initialisations, and a random state of 42.

StandardScaler scales the input features before clustering. This is important because income and spending score have different numerical ranges.

4.3 Model Saving

The trained model package is saved as customer_segmentation_model.pkl using Joblib. The application loads this file to make predictions without retraining the model every time it starts.

The saved package is expected to contain the pipeline, feature names, segment-name mapping, and training ranges.

5. Application Development

The user interface is developed using Streamlit and implemented in app.py.

The application follows these steps:

1. Loads the saved model package.
2. Displays input fields for annual income and spending score.
3. Checks the inputs against the recorded training ranges.
4. Arranges the input values in the correct feature order.

5. Uses the saved pipeline to predict the cluster.
6. Retrieves the corresponding customer segment name.
7. Displays the prediction and submitted customer information.

The interface also provides information about the application and explains that clustering results do not guarantee future customer behaviour.

6. Application Testing

Three sample inputs were previously tested in the application.

Test Case	Annual Income (k$)	Spending Score	Previously Observed Cluster	Previously Displayed Segment
1	15	20	4	Low-Engagement Customers
2	50	50	0	Average-Value Customers
3	100	80	1	High-Value Active Customers

These results were observed during earlier application runs. All three test cases must be rerun after the final code changes, and the actual cluster IDs and segment names must be recorded in the application testing workbook.

A test should be marked as passed only after the actual output has been checked against the expected result. Segment names must also be verified against the saved model's cluster characteristics.

7. Results and Discussion

The application successfully loaded the saved pipeline and returned cluster predictions for the sample customer inputs. The cleaned interface displays the predicted cluster ID, customer segment name, annual income, and spending score.

The results demonstrate how a trained clustering model can be integrated into a user-facing application. However, the cluster IDs themselves do not indicate whether a customer is low-value, average-value, or high-value. These interpretations depend on the actual cluster centres and the segment-name mapping.

8. Limitations

K-Means clustering groups customers according to similarity in the selected features. The output does not establish a causal relationship or guarantee future purchasing behaviour.

Predictions may be less reliable for inputs outside the training ranges. In addition, customer segment descriptions should be validated against the characteristics of each cluster to avoid incorrect interpretations.

9. Conclusion

The Mall Customer Segmentation project demonstrates the use of K-Means clustering in an end-to-end machine learning application. The project includes a Jupyter Notebook, a saved model package, and a Streamlit interface that accepts customer inputs and returns cluster predictions.

The application provides a practical demonstration of customer grouping based on annual income and spending score. Final testing and verification of the segment-name mapping are necessary to ensure the results are presented accurately.